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Lucia Faravelli, Editor
Structural Control and Health Monitoring
Wiley

Dear Editor,

Submission of an original research article: “Load observability of continuous girder bridges: a general response-integral theorem, optimal offset-immune estimation and field verification”

We submit the above manuscript for consideration as a Research Article in *Structural Control and Health Monitoring*.

Influence-area bridge weigh-in-motion, introduced by Ojio and Yamada, estimates the gross weight of a vehicle from the area under one response record. On continuous girders the influence line changes sign over the interior supports, and positive and negative areas partly cancel. The manuscript gives criteria for designing load-monitoring channels on such girders: which channel, at which position, can observe the total load, how well, and how robustly to uncertain boundary conditions.

The paper makes the following contributions.

1. A theorem for any linear response channel: over a complete quasi-static crossing, the record integral equals the total load times the channel’s response to a uniform unit load. It gives closed-form blind loci for deflection, strain, rotation and reaction channels. Strain gauges are blind at every zero of the uniform-load moment diagram, which every continuous girder has.
2. Observability and conditioning indicators that remain valid on haunched girders: a strain indicator that needs no reference stiffness, and a cancellation ratio that separates cancellation between spans from restraint.
3. The minimum-variance invariant multi-channel estimator, an offset-immune variant with zero-sum weights that removes any common-mode offset or drift, and a proof that optimal sensors lie at the extremes of the uniform-load response.

4. An identifiability analysis of the indicator under uncertain pier stiffness, which shows which additional measurement identifies it.

5. Verification on a laboratory two-span beam and on public moving-truck strain records with weighed trucks from a five-span continuous box girder. There, a 5-axle and a 3-axle truck gave the same constant, the integral was speed-independent, superposition held, and gauges at one section had constants that a beam model cannot predict. An application to an in-service 55 + 90 + 55 m box girder shows that its side-span deflection gauge is not identifiable from static acceptance data, and predicts the measurements that would decide.

We believe the paper fits the journal's interest in structural health monitoring theory, sensor placement, signal processing and field measurement. The journal has recently published work on deflection influence lines of continuous beam bridges, which we cite and distinguish.

Data availability. The field strain records are from the public NOR-BWIM dataset (figshare, https://doi.org/10.7294/26880898.v1, CC0 licence), which we cite with its documentation. The laboratory data are listed in full in the manuscript. The processed data of the in-service bridge, the derived data behind every figure and table, and all analysis scripts with their console output are provided as supplementary material, and the bridge data will be deposited at [NEED VALUE: repository name and DOI].

The manuscript is original, has not been published, and is not under consideration by any other journal. [NEED VALUE: two related papers by the authors are under review at *Structures*; state here whether they share the laboratory rig, data or text with this manuscript and how they are cited, or confirm that there is no overlap.] All authors have approved the submission. Conflicts of interest, funding and the use of generative AI are declared in the manuscript.

Yours sincerely,

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